

By Theorem 2, the augmented matrix is inconsistent if $b \neq 0$.
 In this case, if $h-5 \neq 0 \Rightarrow \boxed{h \neq 5}$, the system is inconsistent

$\therefore$ If $h=5$, $h-5=0$, there exists no row of the form $[0 \ 0 \ 0 \ \dots \ 0 \ b]$ where $b \neq 0$. By Theorem 2, then the system is consistent, i.e. there exists a solution set.

$\therefore$ The vector $\begin{bmatrix} -4 \\ 3 \\ 3 \end{bmatrix}$ is in the Span of the vectors $\begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}$, $\begin{bmatrix} 5 \\ -4 \\ -1 \end{bmatrix}$ and $\begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix}$. $\square$

3) Let w_1, w_2, w_3, u and v be vectors in $\mathbb{R}^n$. Suppose the vectors u and v are in $\text{Span}\{w_1, w_2, w_3\}$. Show that $u+v$ is also in $\text{Span}\{w_1, w_2, w_3\}$.

Solution: By defn. of Span, both u, v are linear combinations of w_1, w_2 and w_3 . $\therefore x_1 w_1 + x_2 w_2 + x_3 w_3 = u$ and $y_1 w_1 + y_2 w_2 + y_3 w_3 = v$, where

$$\boxed{x_1, y_1, x_2, y_2, x_3, y_3 \in \mathbb{R}}$$

By vector addition, $u+v = (x_1 w_1 + x_2 w_2 + x_3 w_3) + (y_1 w_1 + y_2 w_2 + y_3 w_3)$
 $= (x_1 + y_1)w_1 + (x_2 + y_2)w_2 + (x_3 + y_3)w_3$

It's easy to see that $(x_1 + y_1), (x_2 + y_2), (x_3 + y_3) \in \mathbb{R}$.

$\therefore$ The vector $u+v$ is a linear combination of w_1, w_2 and w_3 .
 By definition, $(u+v)$ is in the $\text{Span}\{w_1, w_2, w_3\}$.

Exercise 1.3

1. $u = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$ $v = \begin{bmatrix} -3 \\ -1 \end{bmatrix}$. Compute $(u+v)$ and $(u-2v)$

$$u+v = \begin{bmatrix} -1 \\ 2 \end{bmatrix} + \begin{bmatrix} -3 \\ -1 \end{bmatrix} = \begin{bmatrix} -1+(-3) \\ 2+(-1) \end{bmatrix} = \begin{bmatrix} -4 \\ 1 \end{bmatrix}$$

$$u-2v = \begin{bmatrix} -1 \\ 2 \end{bmatrix} - 2 \begin{bmatrix} -3 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} + \begin{bmatrix} -2 \times (-3) \\ -2 \times (-1) \end{bmatrix} = \begin{bmatrix} -1+6 \\ 2+2 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$$